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Machine Learning

Assignment 4

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Abstract

This assignment explores fundamental concepts in unsupervised and supervised machine learning through theoretical analysis and practical experimentation. Topics include K-Means clustering, hierarchical agglomerative clustering, dimensionality reduction using PCA, Kernel PCA, and Linear Discriminant Analysis, as well as feature selection methods spanning filter, wrapper, and embedded approaches. Analytical exercises examine algorithmic behavior, optimization properties, and theoretical limits, while implementation-based tasks apply these methods to real and synthetic datasets. Experiments on the Iris dataset, nonlinear concentric circles, and a heart disease dataset highlight the strengths and limitations of linear and nonlinear techniques for dimensionality reduction and classification. The assignment concludes with image segmentation using K-Means and Gaussian Mixture Models, demonstrating the impact of feature design and probabilistic modeling on clustering outcomes. Overall, the results emphasize the importance of aligning model assumptions with data geometry and learning objectives.

Question 1 — One Full Iteration of K-Means

We are given a dataset of five two-dimensional points and two initial cluster centroids:

$$\mu_0 = (5.3, 3.5), \quad \mu_1 = (5.1, 4.2)$$

Using Euclidean distance, each data point is assigned to its nearest centroid.

After computing distances, the resulting cluster assignments are:

- Cluster 0: (5.5, 3.1), (6.6, 3.0), (6.8, 3.8)
- Cluster 1: (5.1, 4.8), (5.5, 4.6)

The updated centroids are computed as the mean of the points in each cluster. For Cluster 0:

$$\mu_0^{\text{new}} = \left(\frac{5.5 + 6.6 + 6.8}{3}, \frac{3.1 + 3.0 + 3.8}{3} \right) = (6.30, 3.30)$$

For Cluster 1:

$$\mu_1^{\text{new}} = \left(\frac{5.1 + 5.5}{2}, \frac{4.8 + 4.6}{2} \right) = (5.30, 4.70)$$

The final cardinality of Cluster 0 is 3.

Question 2 — Linkage Strategies in Agglomerative Clustering

Computational Complexity

Both single linkage and complete linkage agglomerative clustering algorithms have the same asymptotic computational complexity. In a naive implementation, the time complexity is $\mathcal{O}(n^3)$ due to repeated updates of inter-cluster distances after each merge, while the space complexity is $\mathcal{O}(n^2)$ for storing the pairwise distance matrix. The primary difference between the two methods lies in how the distance between clusters is defined rather than their computational cost.

Robustness to Outliers

Complete linkage is more robust to outliers compared to single linkage. Single linkage defines the distance between two clusters as the minimum distance between their members, which can lead to the chaining effect, where clusters are merged due to a single close pair of points. In contrast, complete linkage uses the maximum distance between cluster members, causing outliers to increase inter-cluster distances and delay premature merging.

Complete Linkage Clustering and Dendrogram

We apply bottom-up agglomerative clustering with complete linkage to the following nine points:

$$\begin{aligned} p_1 &= (1, 1), p_2 = (3, 2), p_3 = (9, 1), \\ p_4 &= (3, 7), p_5 = (7, 2), p_6 = (9, 7), \\ p_7 &= (4, 8), p_8 = (8, 3), p_9 = (1, 4) \end{aligned}$$

The Euclidean distance metric is used:

$$d(p_i, p_j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$

Initially, each point forms its own cluster. At each iteration, the two clusters with the smallest maximum pairwise distance (complete linkage criterion) are merged.

The earliest merges occur between p_4 and p_7 , and between p_5 and p_8 , both at distance $\sqrt{2}$. Points p_1 and p_2 merge next at distance $\sqrt{5}$. Subsequently, point p_3 joins the cluster $\{p_5, p_8\}$, while point p_9 joins the cluster $\{p_1, p_2\}$. Point p_6 merges later with $\{p_4, p_7\}$ due to larger maximum distances.

The final merges occur at higher distance levels, reflecting the compact-cluster preference of complete linkage. The full hierarchical structure is visualized in the dendrogram shown in Figure 1.

Question 3 — Dimensionality Reduction and Data Geometry

Different dimensionality reduction techniques are suitable for different data geometries and learning objectives.

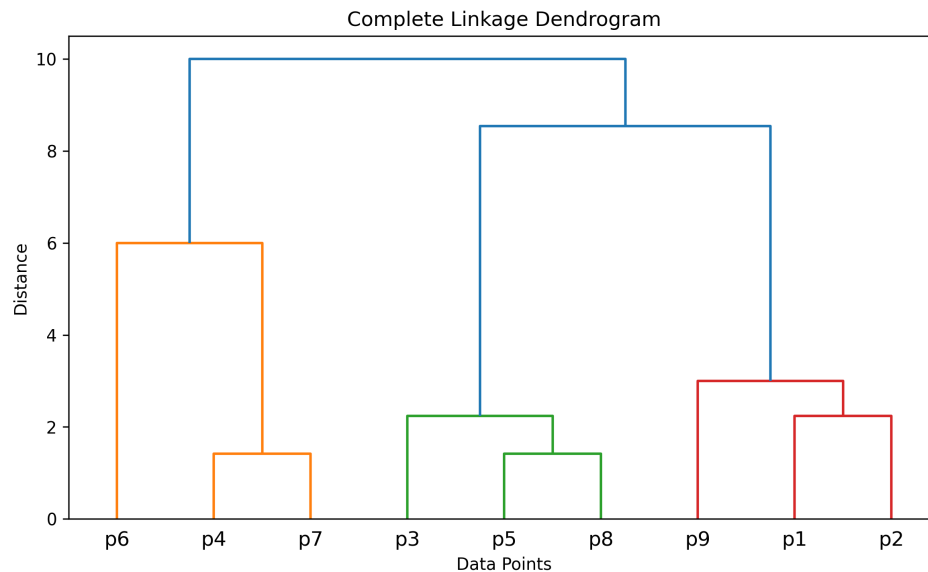


Figure 1: Dendrogram obtained using complete linkage agglomerative clustering. The merge heights correspond to the maximum inter-cluster Euclidean distances.

Linearly Correlated Gaussian Data

Principal Component Analysis (PCA) is the most appropriate method for linearly correlated Gaussian data. PCA maximizes the variance of the projected data and effectively captures the dominant linear correlations present in Gaussian distributions. Since the data structure is linear, kernel-based methods provide no additional benefit.

Two Concentric Circles

Kernel PCA is the most suitable method for datasets consisting of two concentric circles. Such data is not linearly separable, and standard PCA fails to uncover the underlying structure. By applying a nonlinear mapping through a kernel function, Kernel PCA can represent the data in a feature space where the classes become linearly separable.

High-Dimensional Data with Strong Class Overlap

Linear Discriminant Analysis (LDA) is preferred for high-dimensional data with strong class overlap. LDA explicitly uses class labels and seeks to maximize the ratio of between-class variance to within-class variance, leading to projections that improve class separability even when variance-based methods fail.

Data with Irrelevant or Noisy Features

When labels are available, LDA is effective at suppressing irrelevant or noisy features by emphasizing discriminative directions. In unsupervised settings, PCA may reduce dimensionality by discarding low-variance noise, although it may retain high-variance but irrelevant features. Kernel PCA is generally unsuitable in this scenario due to its sensitivity to noise.

Question 4 — Formal Theory of LDA and Kernel PCA

Maximum Number of Discriminant Directions in LDA

Consider a classification problem with C classes. The between-class scatter matrix in Linear Discriminant Analysis is defined as

$$S_B = \sum_{c=1}^C n_c (\mu_c - \mu)(\mu_c - \mu)^T$$

where μ_c is the mean of class c , μ is the global mean, and n_c is the number of samples in class c .

Since the class mean vectors satisfy

$$\sum_{c=1}^C n_c (\mu_c - \mu) = 0,$$

the vectors $(\mu_c - \mu)$ are linearly dependent. Consequently, the rank of S_B is at most $C - 1$:

$$\text{rank}(S_B) \leq C - 1$$

LDA solves the generalized eigenvalue problem

$$S_B w = \lambda S_W w$$

where S_W is the within-class scatter matrix. The number of nonzero eigenvalues, and thus the number of discriminant directions, is bounded by the rank of S_B . Therefore, Linear Discriminant Analysis can produce at most $C - 1$ nonzero discriminant directions.

Kernel PCA Formulation

In Kernel PCA, data points are mapped into a feature space $\mathcal{F}$ via a nonlinear mapping $\phi(\cdot)$. The covariance operator in feature space is given by

$$C = \frac{1}{N} \sum_{i=1}^N \phi(x_i) \phi(x_i)^T$$

Direct eigen-decomposition of C is infeasible when the feature space is high-dimensional or infinite-dimensional.

Assuming that eigenvectors lie in the span of the mapped training samples, we write

$$v = \sum_{i=1}^N \alpha_i \phi(x_i)$$

Substituting into the eigenvalue equation yields the kernel eigenvalue problem

$$K\alpha = N\lambda\alpha$$

where the kernel matrix K has entries $K_{ij} = k(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle$.

The projection of a new data point x onto the k -th principal component is given by

$$y_k(x) = \sum_{i=1}^N \alpha_i^{(k)} k(x_i, x)$$

which depends only on kernel evaluations.

Question 5 — Manual Computation of LDA

We apply Linear Discriminant Analysis (LDA) to a synthetic dataset consisting of two classes, each with five two-dimensional samples.

Dataset

Class 1 consists of the following points:

$$(4, 2), (2, 4), (2, 3), (3, 6), (4, 4)$$

Class 2 consists of:

$$(9, 10), (6, 8), (9, 5), (8, 7), (10, 8)$$

Class Means and Overall Mean

The sample mean of Class 1 is

$$\mu_1 = \frac{1}{5}(4 + 2 + 2 + 3 + 4, 2 + 4 + 3 + 6 + 4) = (3, 3.8)$$

The sample mean of Class 2 is

$$\mu_2 = \frac{1}{5}(9 + 6 + 9 + 8 + 10, 10 + 8 + 5 + 7 + 8) = (8.4, 7.6)$$

The overall mean of all samples is

$$\mu = (5.7, 5.7)$$

Within-Class Scatter Matrix

The within-class scatter matrix is defined as

$$S_W = \sum_{c=1}^2 \sum_{x \in c} (x - \mu_c)(x - \mu_c)^T$$

For Class 1:

$$\sum_{x \in C_1} (x - \mu_1)(x - \mu_1)^T = \begin{bmatrix} 6 & -4 \\ -4 & 10.8 \end{bmatrix}$$

For Class 2:

$$\sum_{x \in C_2} (x - \mu_2)(x - \mu_2)^T = \begin{bmatrix} 9.2 & 1.4 \\ 1.4 & 0.4 \end{bmatrix}$$

Thus,

$$S_W = \begin{bmatrix} 15.2 & -2.6 \\ -2.6 & 11.2 \end{bmatrix}$$

Between-Class Scatter Matrix

The between-class scatter matrix is defined as

$$S_B = \sum_{c=1}^2 n_c (\mu_c - \mu)(\mu_c - \mu)^T$$

With $n_1 = n_2 = 5$:

$$S_B = 5(\mu_1 - \mu)(\mu_1 - \mu)^T + 5(\mu_2 - \mu)(\mu_2 - \mu)^T$$

This yields

$$S_B = \begin{bmatrix} 72.9 & 51.3 \\ 51.3 & 36.1 \end{bmatrix}$$

Optimal Discriminant Direction

For the two-class case, the Fisher optimal direction is given by

$$w = S_W^{-1}(\mu_2 - \mu_1)$$

We have

$$\mu_2 - \mu_1 = (5.4, 3.8)$$

and

$$S_W^{-1} = \frac{1}{163.8} \begin{bmatrix} 11.2 & 2.6 \\ 2.6 & 15.2 \end{bmatrix}$$

Therefore,

$$w \propto (0.46, 0.47)$$

Any scalar multiple of w represents the same discriminant direction.

Projection of Data Points

Each data point is projected onto the LDA direction using

$$y = w^T x$$

Using $w = (0.46, 0.47)$, the one-dimensional projections are:

Class 1 Projections

$$(4, 2) \rightarrow 2.78$$

$$(2, 4) \rightarrow 2.80$$

$$(2, 3) \rightarrow 2.33$$

$$(3, 6) \rightarrow 4.20$$

$$(4, 4) \rightarrow 3.72$$

Class 2 Projections

$$(9, 10) \rightarrow 8.87$$

$$(6, 8) \rightarrow 6.52$$

$$(9, 5) \rightarrow 6.47$$

$$(8, 7) \rightarrow 6.95$$

$$(10, 8) \rightarrow 8.36$$

The projected values demonstrate a clear separation between the two classes along the LDA direction.

Question 6 — Feature Selection and Greedy Methods

Sequential Feature Selection Methods

Sequential Forward Selection (SFS) is a wrapper-based feature selection method that starts with an empty feature set. At each iteration, the feature whose addition leads to the greatest improvement in model performance is added. The algorithm terminates when performance no longer improves or when a predefined number of features is selected.

Sequential Backward Selection (SBS) operates in the opposite direction. It starts with the full set of features and iteratively removes the feature whose removal causes the smallest degradation in performance. The process continues until a stopping criterion is met.

Bidirectional (stepwise) selection combines elements of both SFS and SBS by allowing forward feature additions followed by optional backward eliminations. While more flexible, it still relies on local decisions.

These methods are considered greedy because they make locally optimal choices at each step without revisiting previous decisions.

Consequences of Greedy Behavior

The greedy nature of sequential selection methods implies that they are not guaranteed to find the globally optimal feature subset. Feature selection is a combinatorial optimization problem, and greedy algorithms explore only a limited subset of all possible feature combinations.

For example, two features may individually appear uninformative but become highly discriminative when used together. Sequential Forward Selection may reject both features early, preventing the discovery of their

joint contribution. Similarly, Sequential Backward Selection may remove a critical feature prematurely.

Comparison with RFE and L1-Regularized Methods

Recursive Feature Elimination (RFE) iteratively removes features based on model coefficients, providing a more global view of feature importance than sequential selection methods. However, RFE remains sensitive to feature correlations and can be computationally expensive.

Embedded feature selection using L1 regularization, such as Lasso, incorporates feature selection directly into the model training process by promoting sparse solutions. While computationally efficient and suitable for high-dimensional datasets, L1 regularization tends to select one feature arbitrarily among correlated groups and may fail to capture feature interactions.

Overall, while sequential selection methods are intuitive and simple, they are limited by their greedy nature and are often outperformed by embedded or elimination-based approaches in complex settings.

Question 7(a) — PCA on the Iris Dataset

In this experiment, Principal Component Analysis (PCA) is applied to the Iris dataset in order to reduce the dimensionality from four features to two while preserving as much variance as possible.

PCA Computation and Explained Variance

The Iris dataset consists of 150 samples with four numerical features. Prior to applying PCA, all features are standardized to have zero mean and unit variance. The covariance matrix of the standardized data is then computed, followed by eigen-decomposition.

The resulting explained variance ratios for the four principal components are:

$$[0.7296, 0.2285, 0.0367, 0.0052]$$

The first principal component captures approximately 72.96% of the total variance, while the second principal component captures approximately 22.85%. Together, the first two components explain more than 95% of the total variance, justifying the reduction to a two-dimensional representation.

Visualization in PCA Space

The standardized data is projected onto the first two principal components. Figure 2 shows the resulting two-dimensional representation, where different colors correspond to the three Iris classes.

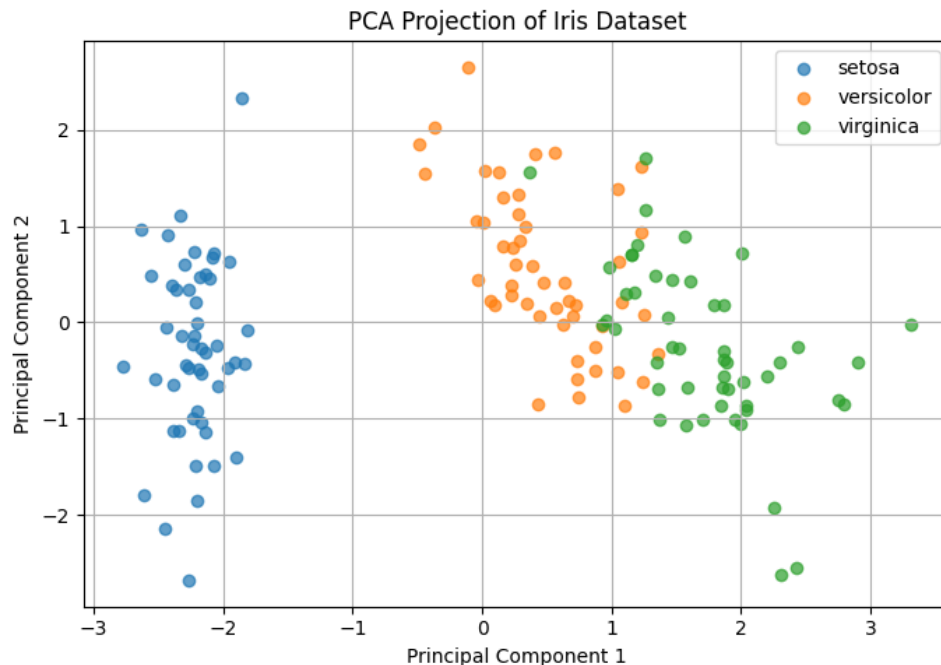


Figure 2: Two-dimensional PCA projection of the Iris dataset. Each point represents a sample, colored according to its class label.

Analysis of Results

The first principal component captures the largest amount of variance, which can be intuitively explained by the strong variability and correlation present in petal-related features such as petal length and petal width. PCA aligns the first component with this dominant direction of variance.

In the projected space, the Setosa class is clearly separated from the other two classes, while Versicolor and Virginica exhibit noticeable overlap. This behavior is expected because PCA does not use class labels and optimizes only for variance preservation rather than class separability.

Suitability of PCA for Classification

PCA is not necessarily an optimal dimensionality reduction technique when the primary goal is classification. Since PCA ignores class labels,

the directions of maximum variance may not correspond to the most discriminative directions. As a result, class overlap may persist even after dimensionality reduction.

If Linear Discriminant Analysis (LDA) were applied instead of PCA, the projection would be fundamentally different. LDA explicitly incorporates class labels and seeks to maximize the ratio of between-class variance to within-class variance. Consequently, LDA would aim to project the data into a space where class separability is maximized, likely resulting in improved separation between Versicolor and Virginica compared to PCA.

Question 7(b) — PCA vs Kernel PCA on a Nonlinear Dataset

In this experiment, the limitations of linear dimensionality reduction methods are examined using a nonlinear dataset consisting of two nested circles.

Dataset Description

A two-dimensional dataset with two nested (concentric) circles is generated using the `sklearn.datasets.make_circles` function. One class forms an inner circle, while the other forms an outer circle, resulting in a dataset that is not linearly separable. Figure 3 illustrates the original data distribution.

Standard PCA Projection

Principal Component Analysis is applied to the standardized data and the projection onto the first two principal components is computed. The explained variance ratios of the two components are approximately

$$[0.5003, 0.4997],$$

indicating that the variance is almost equally distributed between the two components due to the symmetry of the dataset.

Figure 4 shows the PCA projection. The two classes remain highly overlapped, demonstrating that PCA fails to separate the nested circles. This behavior is expected because PCA is a linear method that preserves variance rather than nonlinear structure or class separability.

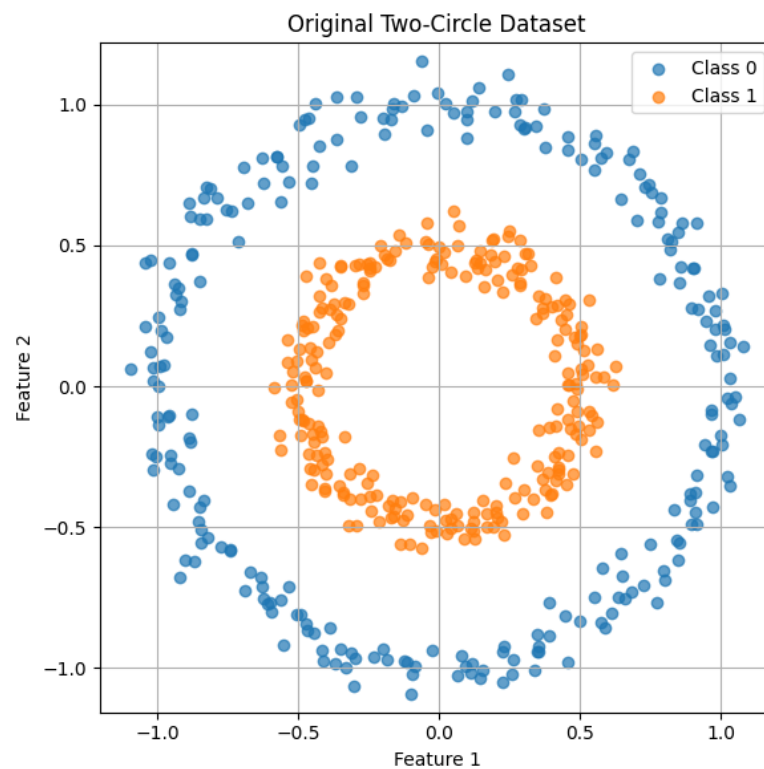


Figure 3: Original two-dimensional dataset consisting of two nested circles.

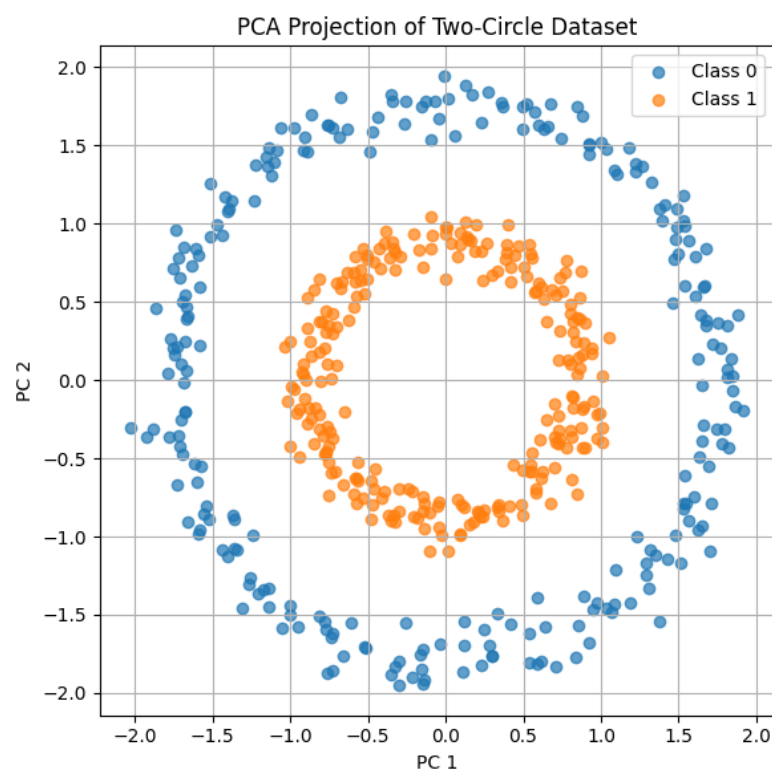


Figure 4: Projection of the two-circle dataset onto the first two principal components using standard PCA.

Kernel PCA Projection

Kernel PCA with a Gaussian (RBF) kernel is then applied to the same dataset. By implicitly mapping the data into a higher-dimensional feature space, Kernel PCA is able to capture the nonlinear structure of the data.

As shown in Figure 5, the two classes become clearly separable in the transformed space. One class collapses near the center, while the other is spread along a nonlinear manifold, enabling effective separation.

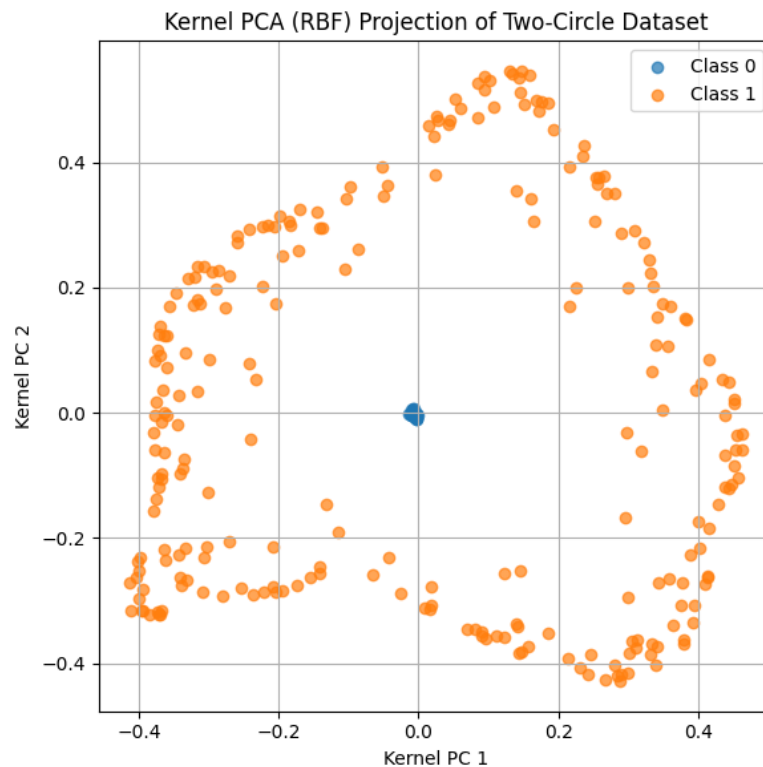


Figure 5: Kernel PCA (RBF) projection of the two-circle dataset, showing clear class separation.

Discussion

Standard PCA fails on this dataset because it is restricted to linear projections and cannot unfold nonlinear manifolds such as concentric circles. In contrast, Kernel PCA leverages the kernel trick to perform nonlinear feature mapping, allowing the intrinsic structure of the data to be revealed. This experiment highlights the importance of choosing dimensionality reduction techniques that align with the geometry of the underlying data.

Question 8 — Feature Selection and Visualization on a Real Dataset

In this experiment, multiple feature selection techniques are applied to the Heart Disease dataset, treated as a binary classification problem. The methods are compared in terms of predictive performance, interpretability, robustness, and sensitivity to feature correlations.

Exploratory Analysis and Correlation-Based Filtering

Feature distributions and a correlation matrix are examined to identify redundancy among input features. Figure 6 shows the correlation matrix. No pair of features exceeds the correlation threshold of 0.85; therefore, no features are removed using correlation-based filtering.

This result indicates that correlation-based filtering may be ineffective when features are weakly correlated or when redundancy is not captured by pairwise linear correlation alone.

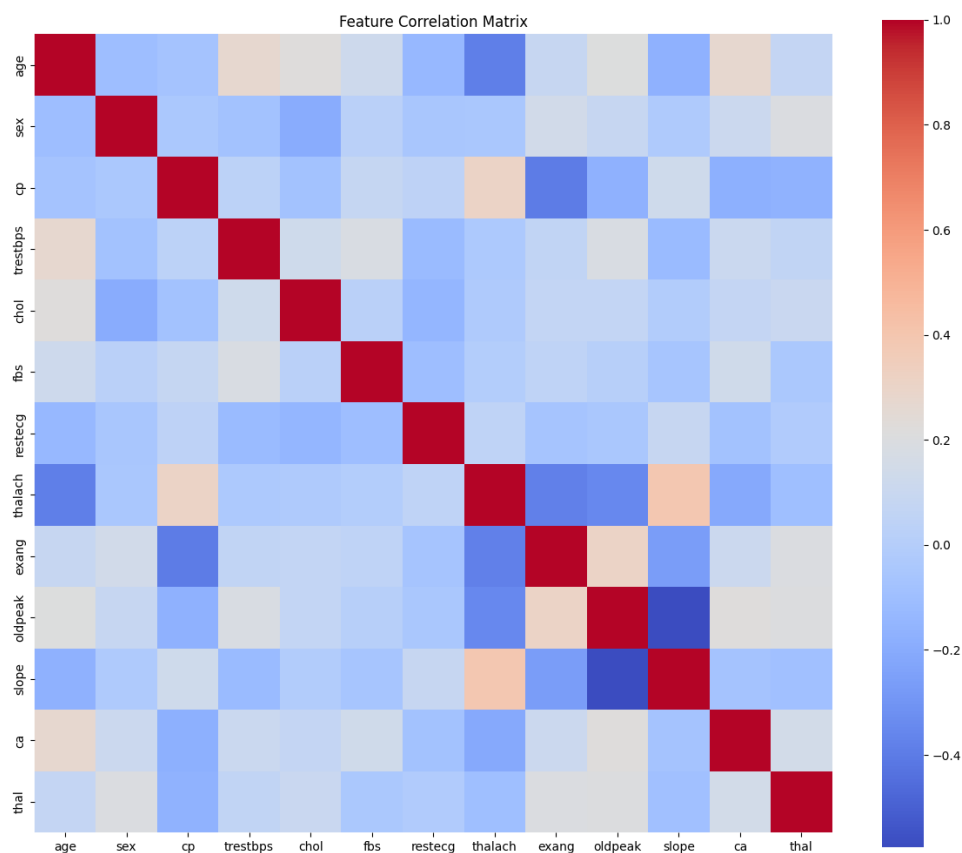


Figure 6: Correlation matrix of Heart Disease dataset features.

Baseline Classification

A logistic regression classifier trained using all 13 features achieves a baseline accuracy of

0.8249.

This serves as a reference point for evaluating feature selection methods.

Univariate Feature Selection

Mutual information is used to rank features based on individual relevance to the target variable. The top five features selected are `chol`, `thalach`, `oldpeak`, `thal`, and `cp`. Training a classifier using only these features results in an accuracy of

0.8132.

The slight performance drop reflects the inability of univariate methods to capture feature interactions.

Recursive Feature Elimination (RFE)

RFE with logistic regression selects five features: `sex`, `cp`, `exang`, `oldpeak`, and `ca`. This model achieves the highest accuracy of

0.8521,

demonstrating the effectiveness of wrapper methods in capturing feature interactions while maintaining a compact feature subset.

RFECV

RFECV selects nine features using cross-validation, achieving an accuracy of

0.8405.

The larger number of selected features indicates limited redundancy in the dataset and highlights the trade-off between sparsity and robustness.

Embedded Feature Selection with L1 Regularization

L1-regularized logistic regression removes only one feature (`fbs`), retaining most features due to correlated and distributed predictive signals. The resulting accuracy is

0.8288,

comparable to the baseline. This illustrates the tendency of L1 regularization to arbitrarily select among correlated features.

PCA as an Exploratory Tool

PCA is applied to analyze variance structure rather than perform feature selection. Approximately 90% of the variance is captured by 10 principal components, and 95% by 11 components, as shown in Figure 7. This confirms that PCA does not provide a compact, interpretable feature subset.

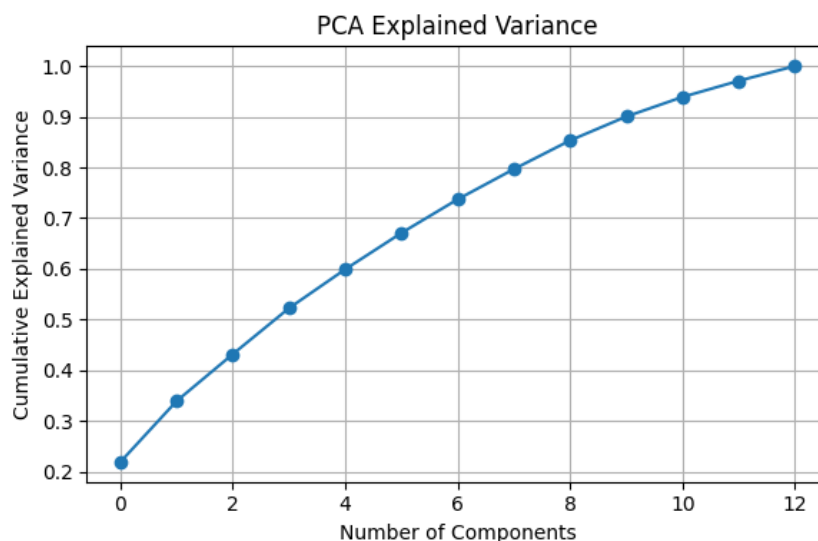


Figure 7: Cumulative explained variance obtained using PCA.

Comparison of Feature Selection Methods

Figure 8 summarizes classification accuracy across all methods. RFE achieves the best performance with the smallest number of features, while RFECV provides a more conservative but stable selection. Univariate and embedded methods show limitations in handling feature interactions and correlations.

Question 9 — Image Segmentation Using K-Means and GMM

In this experiment, image segmentation is performed using K-Means clustering and Gaussian Mixture Models (GMM). Two color images are segmented into $K \in \{2, 3, 4, 5\}$ regions based on both spatial and color information.

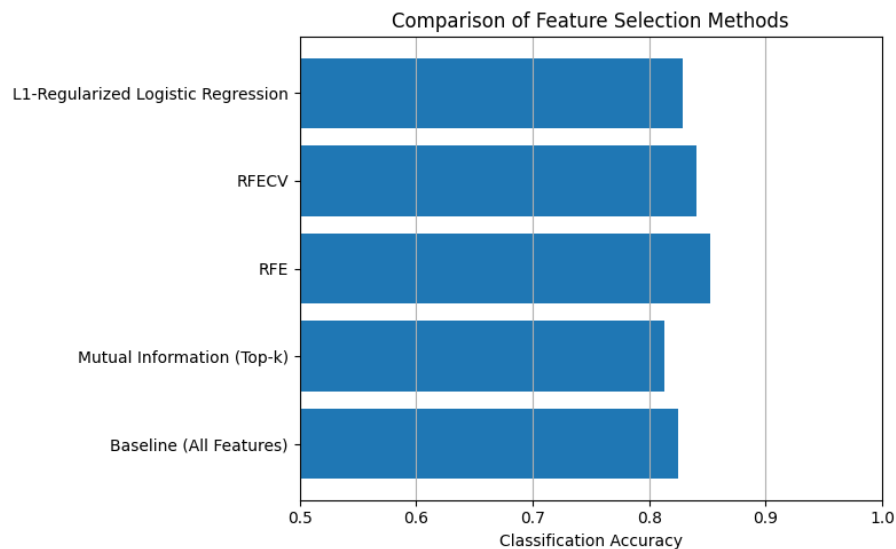


Figure 8: Classification accuracy comparison across feature selection methods.

Original Images

Figure 9 shows the two original color images used for segmentation. Image 1 is shown on the left and Image 2 on the right. All subsequent segmentation results are obtained from these images using identical feature representations and clustering procedures.



Figure 9: Original input images used for segmentation. Image 1 (left) and Image 2 (right).

Feature Representation and Normalization

Each pixel is represented by a five-dimensional feature vector:

$$(y, x, R, G, B),$$

where x and y denote the horizontal and vertical pixel coordinates relative to the top-left corner of the image, and R , G , and B represent the color intensities. All five features are linearly normalized to the interval $[0, 1]$, ensuring that spatial and color components contribute comparably to the distance computations. As a result, all feature vectors lie within the unit hypercube $[0, 1]^5$.

K-Means Based Segmentation

K-Means clustering with Euclidean distance is applied to the five-dimensional pixel features. For each value of K , every pixel is assigned to the nearest cluster centroid, and the resulting cluster labels are reshaped back into image form. To improve visualization contrast, each cluster label is mapped to a distinct color.

Figures 10 and 11 show the segmentation results for the two images using $K = 2, 3, 4$, and 5. As K increases, finer image structures and regions become distinguishable, while spatial coherence is preserved due to the inclusion of pixel coordinates in the feature vector.

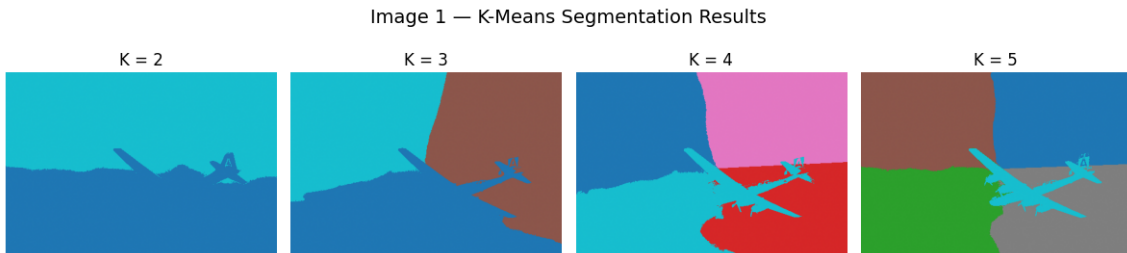


Figure 10: K-Means segmentation results for Image 1 with $K \in \{2, 3, 4, 5\}$.

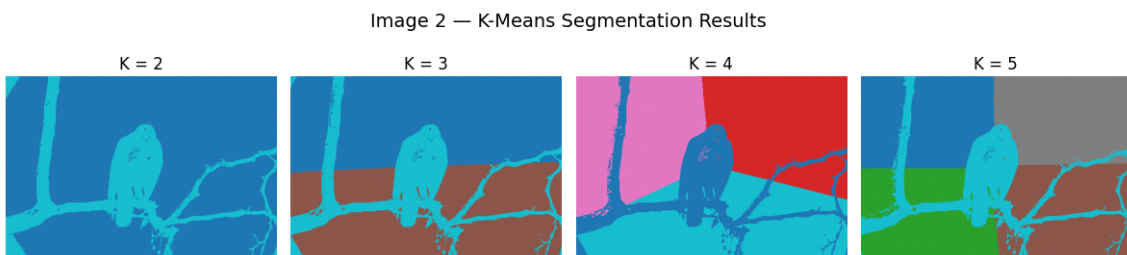


Figure 11: K-Means segmentation results for Image 2 with $K \in \{2, 3, 4, 5\}$.

GMM-Based Segmentation

Gaussian Mixture Models with full covariance matrices are fitted using the Expectation-Maximization (EM) algorithm. Each pixel is assigned to the

mixture component with the highest posterior probability, corresponding to a MAP classification.

Figures 12 and 13 present the GMM-based segmentation results. Compared to K-Means, GMM allows clusters with elliptical shapes in feature space and supports soft probabilistic modeling, often producing smoother and more adaptive segment boundaries.



Figure 12: GMM segmentation results for Image 1 with $K \in \{2, 3, 4, 5\}$.

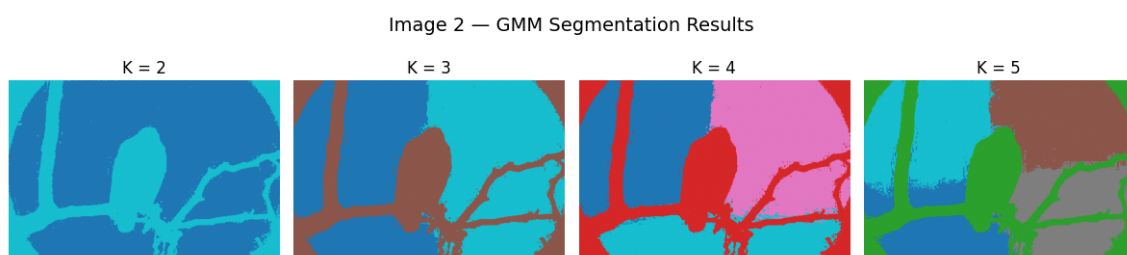


Figure 13: GMM segmentation results for Image 2 with $K \in \{2, 3, 4, 5\}$.

Comparison and Discussion

K-Means produces hard assignments and assumes spherical clusters in feature space, leading to sharp region boundaries. In contrast, GMM models full covariance structures and assigns pixels based on posterior probabilities, which can better capture gradual transitions in color and texture. Differences between the two methods arise from these modeling assumptions, particularly in regions with smooth color gradients or complex spatial structures.